

MAHMOUD MOHAMED LOTFY

Demand Planning & Inventory Optimization Specialist | Forecasting Models | S&OP | Material Continuity

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PROFESSIONAL SUMMARY

Demand Planning and Inventory Optimization Specialist with 15+ years of professional experience bridging operational volume forecasting, historical consumption modeling, and Material Requirements Planning (MRP). Designed and deployed a procedure-linked demand forecasting model that cut emergency procurement by 40–60% while maintaining 100% stock availability across 3 consecutive fiscal years within fixed operational budgets. Maintained zero stockouts and zero procedure cancellations across 30+ consecutive months in a high-complexity cardiac catheterization environment (90–140 procedures/month). Expert in time-series consumption analysis, Safety Stock (SS) formulas, Reorder Point (ROP) planning, min-max thresholds, and eliminating expiry write-offs via FEFO (First-Expired, First-Out) protocols. Highly skilled in building mathematical models, dynamic forecast sheets, and inventory control tools in Advanced Excel (XLOOKUP, INDEX/MATCH, Pivot Tables) and custom software. Proven cross-sector track record spanning healthcare operations (Al-Obour Hospital) and FMCG high-turnover food manufacturing (Verona Sweets).

KEY ACHIEVEMENTS

- Built a procedure-linked demand forecasting model mapping scheduled procedure types to consumable BOMs, slashing emergency purchasing by **40–60%** and achieving **>92%** forecast accuracy on critical consumables
- Maintained **100% stock availability** for critical clinical consumables across **3 consecutive fiscal years** within fixed operating budgets
- Achieved **zero stockouts** and **zero supply-failure procedure cancellations** across **30+ consecutive months (>1,000 procedures)**
- Designed and coded a custom online Inventory Management Application (**~320 development hours**) featuring real-time depletion tracking, barcode scanning, and automated reorder alerts
- Formulated dynamic Safety Stock (SS) and Reorder Point (ROP) equations incorporating supplier lead-time variability and procedure volatility
- Reduced inventory write-offs and consumable expiration losses to near-zero by instituting rigorous FEFO stock rotation protocols
- Managed demand planning and replenishment for 90–140 complex clinical procedures per month with **~100% material readiness**
- Successfully absorbed **70%+** operational cost inflation while keeping inventory carrying costs below budget through disciplined stock turnover
- Built full demand forecasting and inventory tracking infrastructure on **zero external software budget** using Advanced Excel and Google Sheets

SKILLS

- Demand Planning & Forecasting:** Demand Forecasting, Demand Modeling, Time-Series Consumption Analysis, Sales and Operations Planning (S&OP), Material Requirements Planning (MRP), Bill of Materials (BOM) Mapping, Capacity Planning, Run-Rate Tracking, Forecast Error Minimization
- Inventory Optimization & Control:** Safety Stock (SS) Modeling, Reorder Point (ROP) Calculations, Min-Max Inventory Levels, Stockout Prevention, Days of Supply (DOS), FEFO/FIFO Stock Rotation, Expiry Risk Monitoring, Carrying Cost Optimization, Consignment Stock Management
- Analytical Systems & Tooling:** Advanced Microsoft Excel (XLOOKUP, INDEX/MATCH, Dynamic Arrays, Pivot Tables, Scenario Modeling), Google Sheets, Custom Inventory Software, Barcode Tracking Systems, Data Visualization, Performance Dashboards
- Operational Integration & Continuity:** Supply Chain Integration, Budget-Constrained Replenishment, Lead-Time Variance Management, Cross-Functional Alignment, Clinical and Production Staging, Continuous Improvement

PROFESSIONAL EXPERIENCE

Healthcare Operations and Supply Chain Lead | Cath Lab

January 2023 – April 2026

Al-Obour Hospital - Zagazig, Egypt

Led demand planning, inventory optimization, and material continuity for the Cardiac Catheterization Unit processing 90–140 complex procedures per month. Accountable for forecasting accuracy, stock availability, working capital efficiency, and uninterrupted case execution.

- Modeled clinical material requirements by mapping scheduled procedure mixes (diagnostic, PCI, stenting, pacemaker implantations) to specific consumable Bills of Materials (BOM), driving down emergency purchasing by **40–60%** and achieving **>92%** forecast accuracy on critical consumables
- Maintained **100% stock availability** across high-value consumables (stents, balloon catheters, guide wires, introducers) across **3 consecutive fiscal years** within fixed monthly allocations
- Built and deployed a custom online Inventory Management Application (**~320 development hours**) delivering real-time stock balances, barcode scanning, automated consumption tracking, and proactive expiry alerts
- Instituted dynamic Safety Stock (SS) and Reorder Point (ROP) formulas that balanced demand variability against supplier lead-time fluctuations, maintaining **zero stockouts** for **30+ consecutive months**
- Implemented FEFO (First-Expired, First-Out) stock rotation workflows and monthly shelf-life audits, reducing expired product losses to near-zero
- Conducted monthly demand-supply reconciliation reviews with lead cardiologists and procurement teams, aligning procedure scheduling with replenishment lead times (S&OP alignment)
- Applied budget-constrained replenishment ordering with clinical priority ranking to absorb **70%+** inflation while protecting cash flow
- Structured unit inventory tracking dashboards using Advanced Excel, creating automated alerts for low-stock triggers, buffer depletion, and slow-moving items
- Reorganized recovery room and in-OR inventory staging: established standardized stock par-levels (capped at 12 patients in recovery) to eliminate material hoarding and stock shrinkage

Accountant & Unit Coordinator | Cardiac Catheterization Unit

October 2019 – January 2023

Al-Obour Hospital - Zagazig, Egypt

Managed Cath Lab front-desk admissions, patient intake ticketing, billing, point-of-service cash collections, and unit treasury. Coordinated consultant procedure scheduling and inter-departmental patient flow with CCU and Operating Theatres, auditing vendor invoices prior to promotion to Operations Lead.

Stores & Factory Manager

2016 – 2017

Verona Sweets Factory

- Managed demand planning, raw material replenishment, and factory warehouse inventory for a high-turnover FMCG sweets manufacturer
- Analyzed production schedule requirements against raw material lead times to establish buffer stock levels and prevent production line stoppages
- Implemented FIFO material movement, minimizing ingredient spoilage and reducing raw material waste across production batches

Customer Care Manager

September 2018 – July 2019

ClinPrime - Healthcare Software

- Managed operational software support and issue triage for **30+ hospital clients**
- Analyzed client issue volume trends, forecasting peak support demand and reallocating technical resources to cut resolution time from **48 hours to 24 hours**

Data Processing Specialist — International Business Services

2015 – 2016

Customer Service Representative, ADSL Technical Support, Team Lead — ETISAL International / Etisalat

2011 – 2012

ADDITIONAL EXPERIENCE

- **Co-Founder and COO** — Share Events & Marketing Services (2013–2014)
- **Co-Founder** — TEDx Zagazig (2013–2014)
- **Organizer** — Cairo Derma International Conference (2012)
- **Organizer and PR Head** — Zagazig University Vascular Surgery Conference (2011)
- **Founder and CEO** — Share Human & Community Development (2017–2020)
- **Co-Founder** — Karmasha (2015–2016)
- **Founder and Creator** — Frobe (2022–2023)
- **Head of Public Relations** — SIFE Organization (Enactus)
- **Media Production Professional** — Freelance (2014–2019)

EDUCATION

B.Sc. Commerce — Accounting | Zagazig University, Egypt

May 2010

LANGUAGES

Arabic: Native / Full Professional Proficiency

English: Full Professional Proficiency (reading, writing, reporting)

TOOLS AND TECHNOLOGIES

Advanced Microsoft Excel (XLOOKUP, INDEX/MATCH, Pivot Tables, Scenario Modeling) | Google Sheets | Custom Online Inventory Management Application (Barcode & Expiry Tracking) | Visual Studio Code

Applied to build demand forecasting engines, safety stock calculators, inventory depletion models, and automated reorder alerts.

PROFESSIONAL DEVELOPMENT

- Self-Study in Demand Planning, Safety Stock Modeling, and Material Requirements Planning (MRP)
- Self-Built Cath Lab Demand Planning and Inventory Optimization Infrastructure using advanced Excel and Google Sheets on zero external budget
- Continuous learning in forecasting accuracy techniques, S&OP synchronization, and multi-echelon inventory control